

The Systems Approach — Course Notes

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Where to write your answers: if you have access to a device, use the **online course notes** instead — you can type your answers into the answer panels and they are saved as you go. Otherwise, write your answers in your **jotter**, with the task and question number beside each answer.



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Section 1: What is a System?

Engineers and designers are in the business of **solving problems**. Whether designing a new bridge, a home electronic device, or a satellite communications system, engineers use a common way of thinking called the **systems approach**.

A **system** is a collection of components that work together to perform a function. All systems — no matter how simple or complex — can be described using the same basic structure:

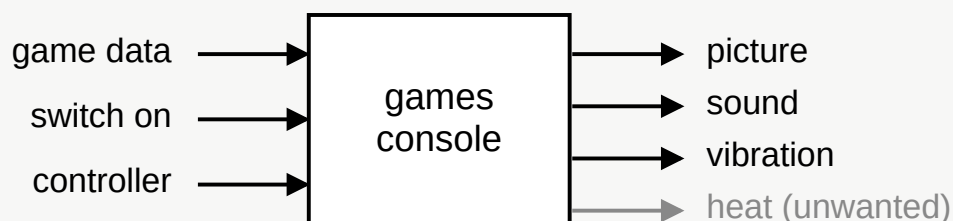
This is called the **universal system diagram**. The word "universal" means it can be used for any situation at all.



- **Input** — what goes into the system to make it work (e.g. electricity, a signal, a physical force)
- **Process** — what the system does to the input
- **Output** — what comes out of the system (desired outputs and unwanted outputs)

WORKED EXAMPLE — SYSTEM DIAGRAM: GAMES CONSOLE

A games console receives game data (from a disc or Wi-Fi connection), is switched on by the user, and receives input from a games controller. It outputs picture, sound and vibration, along with heat as an unwanted output.



⚠ Do **not** include electricity as an input in systems diagrams — electricity is part of the system, not a real-world input. Instead write the real-world input that triggers the system (e.g. "light", "heat", "signal").

Task 1 — Universal systems diagrams

For each device below, fill in the inputs and outputs (including any unwanted outputs) to complete the system diagram.

INPUTS → **KETTLE** → OUTPUTS (WANTED + UNWANTED)

INPUTS → **TOASTER** → OUTPUTS (WANTED + UNWANTED)

INPUTS → **RADIO** → OUTPUTS (WANTED + UNWANTED)

INPUTS → **WASHING MACHINE** → OUTPUTS (WANTED + UNWANTED)

INPUTS → **3D PRINTER** → OUTPUTS (WANTED + UNWANTED)

INPUTS → **FIRE ALARM** → OUTPUTS (WANTED + UNWANTED)

Section 2: Sub-Systems Diagrams

The universal system diagram tells us what goes in and what comes out — but not how it works inside. The next step is to **divide the process box** into its sub-systems, showing how each part of the system interacts.

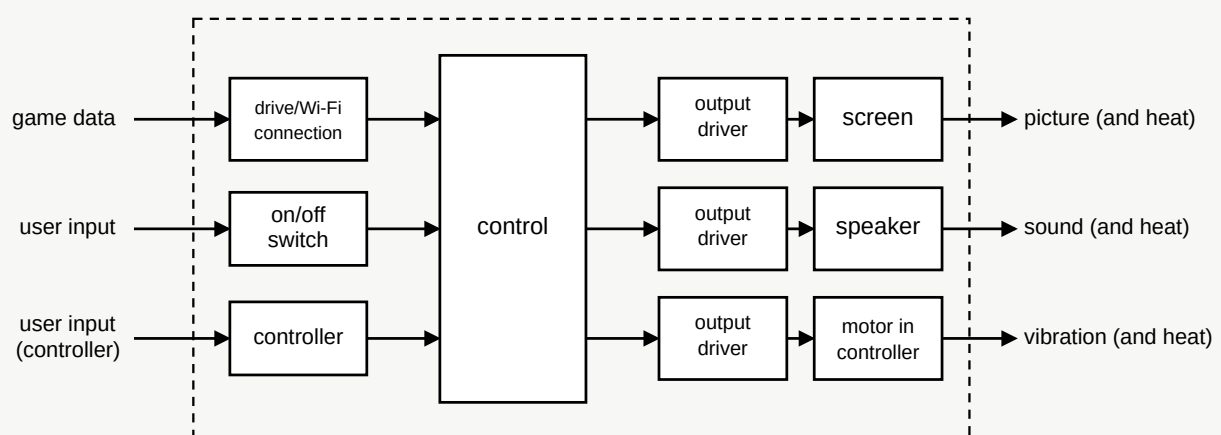
A **sub-systems diagram** shows all the individual components inside the system boundary, each in its own box, with arrows showing how signals or energy flows between them.

Key terms

- **System boundary** — a dashed line enclosing all the sub-systems. Real-world inputs and outputs are outside the boundary.
- **Input device (sensor)** — converts a real-world input into an electrical signal. Examples: thermistor, microphone, switch, light sensor.
- **Output device (actuator)** — converts an electrical signal into a real-world output. Examples: motor, loudspeaker, LED, heater.
- **Output driver** — an electronic component that boosts a low-power control signal from the control unit to provide the higher current or voltage required to operate an output device such as a motor or heating element. Every output device needs a driver (e.g. transistor, relay).

WORKED EXAMPLE — SUB-SYSTEMS DIAGRAM FOR A GAMES CONSOLE

The games console has three real-world inputs: game data (from a disc or Wi-Fi connection), user input via the on/off switch, and user input via the games controller. The outputs are picture and sound (each accompanied by heat), plus vibration from the motor in the controller.



Task 2 — Sub-systems diagrams

1. Draw a sub-systems diagram for a **mobile phone**. Include: screen, touch pad, control unit, on/off switch, aerial, battery, speaker, microphone, headphone jack, output driver. Show the system boundary and label all connections.
2. Draw a sub-systems diagram for an **electric oven**. Include: output driver, heating element, on/off switch, temperature selector switch, fan selector switch, oven light.
3. For each system below, identify the input device and output device, then describe how the system works by explaining the function of each sub-system in order.
 - a) Toaster — input device: timer switch | output device: heating element
 - b) Greenhouse watering (irrigation) system — input device: moisture sensor | output device: water pump
 - c) Factory press machine — input device: push button | output device: motor / solenoid

Section 3: Open-Loop and Closed-Loop Control

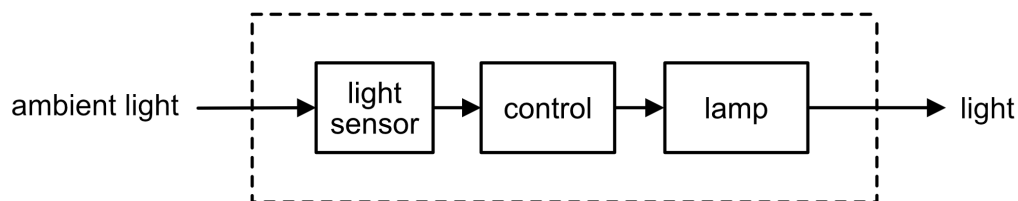
Any system must have some way to **control its operation**. Systems can also be described as **open-loop** or **closed-loop**.

Open-loop control

In an open-loop system, the output simply switches on or off when a certain input happens. There is **no feedback** — the system does not monitor its output. Once activated, it stays on until switched off.

WORKED EXAMPLE — OPEN-LOOP CONTROL: AUTOMATIC STREET LIGHT

An automatic street light uses a light sensor to detect when it gets dark. When the light level drops below a set threshold, the light sensor triggers the control unit, which switches on the LED array via an output driver. There is no feedback — the system cannot detect whether the light is actually illuminating the street. It simply switches on when dark and off when light returns.

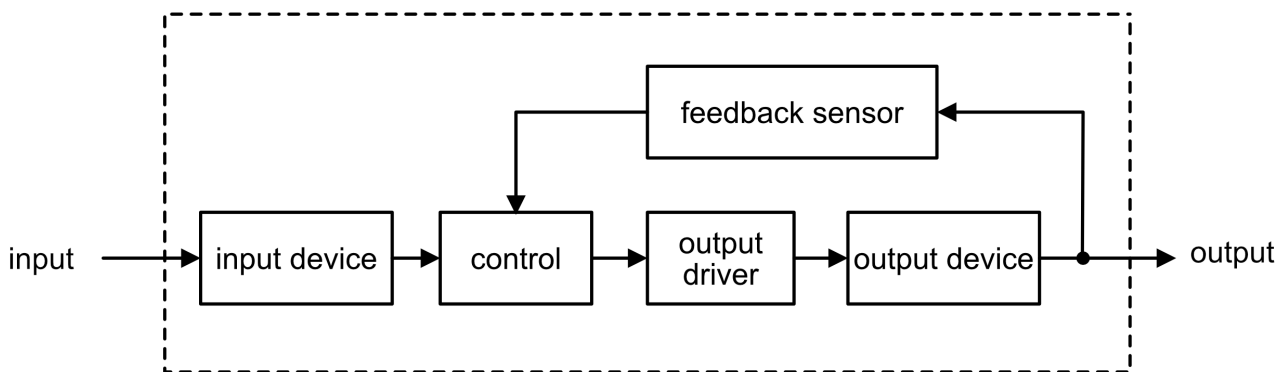


Why do we need feedback?

Imagine an oven with open-loop control: the heater is simply switched on for 20 minutes. On a cold day, with the door opened a few times or a heavy tray inside, the oven ends up cooler than intended. On another day it overshoots. The system has **no way of knowing** what temperature it has actually reached, so it cannot correct itself. To fix that, the system needs to *measure* its own output and act on the result — this is called **feedback**.

Closed-loop control

In a closed-loop system, the output is continually **monitored** and fed back to adjust the process. This makes the system far more accurate. The feedback signal is compared to the desired value and the system corrects itself automatically.



Open-loop systems

- Simple to design and cheap to build
- No feedback — cannot correct errors
- Output may drift from desired value
- Examples: toaster, street lamp timer, microwave oven

Closed-loop systems

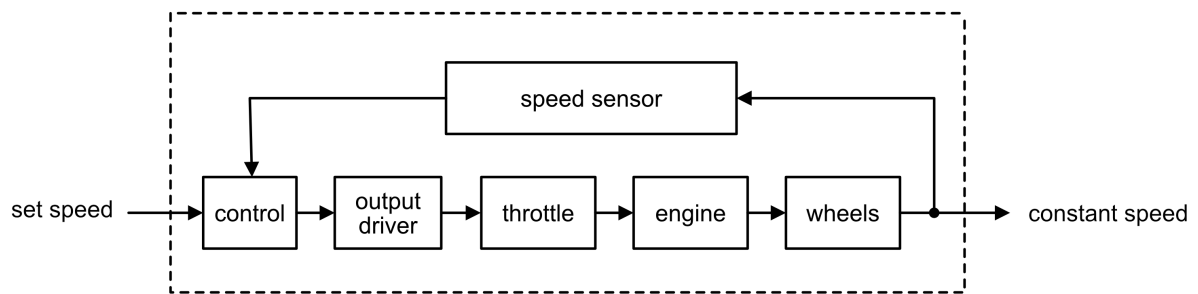
- More complex and expensive
- Feedback allows automatic correction
- Output is maintained at the desired value
- Examples: thermostat, cruise control, hair straighteners with automatic temperature control

Open-loop vs closed-loop: summary

Type	Feedback?	Example
Open-loop	No	Street light activated by a light sensor — switches on when dark, no monitoring of output
Closed-loop	Yes — sensor monitors the output	Car cruise control, central heating thermostat, oven temperature control

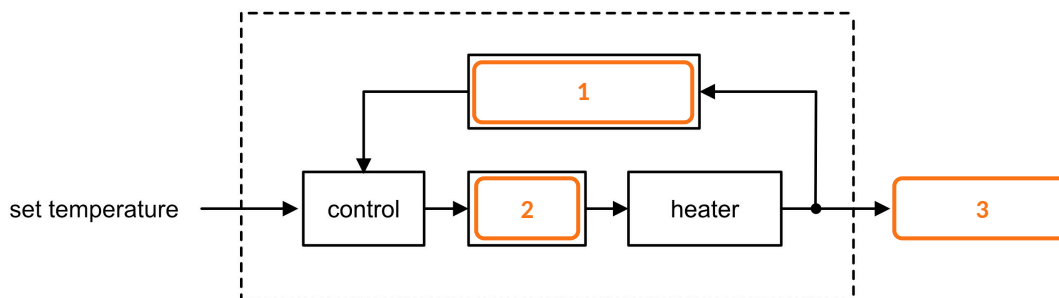
WORKED EXAMPLE — CLOSED-LOOP CONTROL: CAR CRUISE CONTROL

The driver sets a desired speed. A speed sensor (tachogenerator) continuously measures the actual vehicle speed and feeds this back to the control unit. If the actual speed differs from the set speed, the output driver adjusts the throttle to correct it. The system constantly monitors and corrects, keeping the speed accurate.



Task 3 — Open and closed-loop control

1. State one difference between an open-loop and a closed-loop control system.
2. State whether each of the following is open-loop or closed-loop control.
 - a) A microwave oven set to run for 2 minutes.
 - b) Car cruise control that maintains a set speed.
 - c) Hair straighteners with automatic temperature control.
 - d) A security light on a timer that switches on at 6pm every evening.
3. The diagram shows a closed-loop control system for a central heating system. Name the three missing sub-systems (boxes 1, 2 and 3).



 **Exam & assignment tip:** Every line in a sub-systems diagram must have an arrowhead — including both lines into and out of the feedback sensor. The feedback line must always be taken from *after* the output device, not from the device itself.

- a) Name the sub-system that provides the feedback signal.
 - b) What does the feedback signal tell the control unit?
 - c) What action does the control unit take if the room temperature falls below the set value?
4. Draw a closed-loop control diagram for a **refrigerator** that automatically maintains a set temperature. Include: temperature sensor, control unit, output driver, compressor. Show the system boundary and clearly mark arrowheads on both lines of the feedback loop. Remember: the feedback line must come from *after* the output device.

Section 4: Case Study — Domestic Appliances

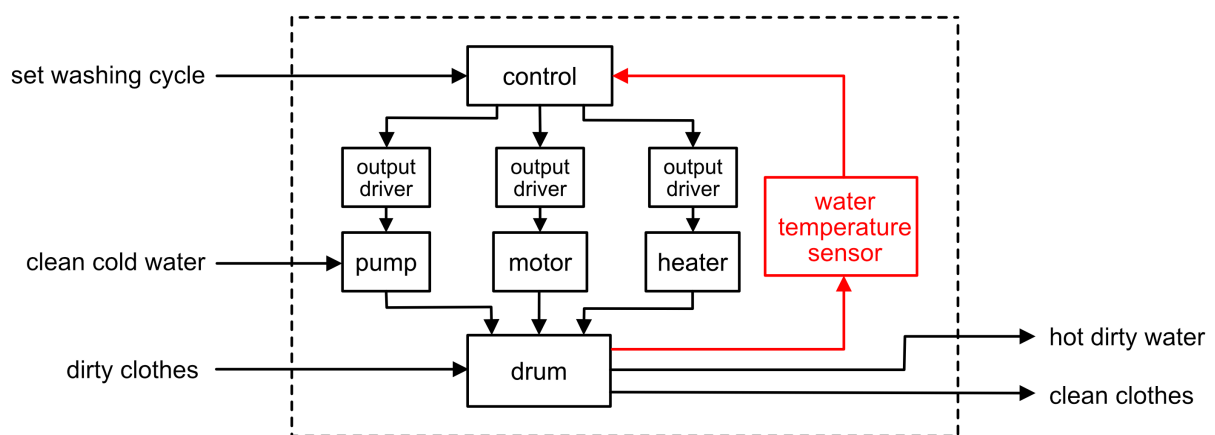
Many everyday appliances use systems principles. The washing machine is a good example of a system with multiple levels of control complexity.

Washing machine

A washing machine is a complex system with both open and closed-loop elements. The wash programme (timer, spin speed) is essentially open-loop — it runs a fixed sequence regardless of outcome. However, the **water heating** uses **closed-loop control** — a temperature sensor monitors the water temperature and feeds back to the control unit, which switches the heater on or off to maintain the correct washing temperature.

WORKED EXAMPLE — CLOSED-LOOP CONTROL: WASHING MACHINE WATER HEATING

The user sets a washing programme and temperature. The control unit switches the heater on via the output driver. A water temperature sensor continuously monitors the water and feeds back to the control unit. When the target temperature is reached, the control unit switches the heater off. If the temperature drops, the heater switches back on. This feedback loop maintains the correct temperature throughout the wash.



Task 4 — Domestic appliances

1a. Sketch a sub-systems diagram for a **clothes iron** with automatic temperature control. Include a temperature selector, control unit, output driver and heating element. Show the system boundary and any feedback path.

1b. Is the iron an open-loop or closed-loop system? Explain your answer with reference to your diagram.

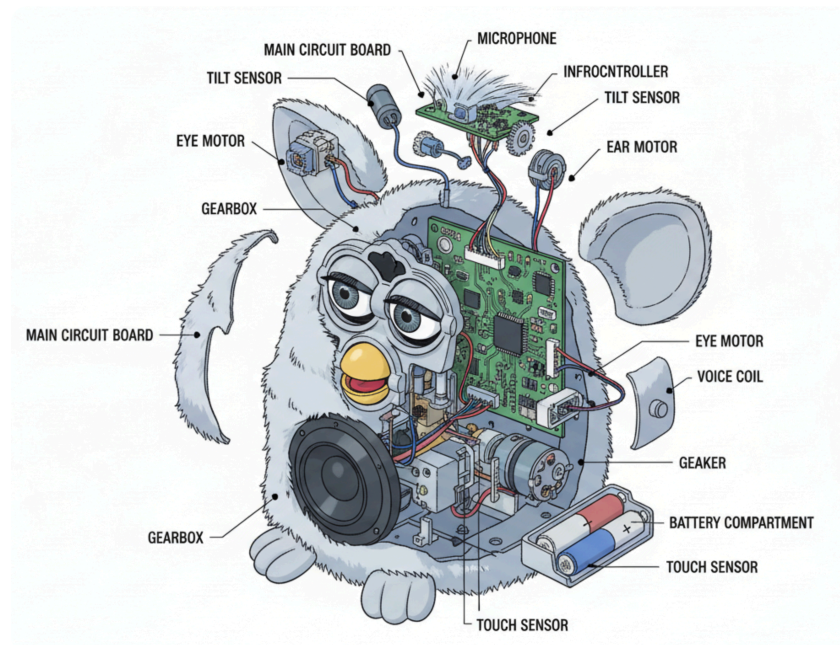
1c. Sketch a sub-systems diagram for a **slow cooker** with a temperature control setting. Show the system boundary, inputs, outputs and all sub-systems inside the boundary.

2a. State the name of the sub-system in the washing machine that provides feedback for temperature control.

2b. Describe, with reference to the sub-systems diagram above, how the water heating system in the washing machine operates. Mention how each sub-system interacts, and the role of feedback.

Section 5: Case Study — The Furby Electronic Toy

The Furby is a good example of a **mechatronic system** — one that uses an electronic circuit to control a number of mechanisms. It contains several sensors so it can react to changes in its environment (e.g. being placed in a dark room or turned upside down).



Task 5 — Furby case study

1. Research the Furby mechatronic toy and briefly describe how it works as a system.
2. List the input devices for the Furby and state the function of each. (Hints: push switches on front and back, switch in mouth, light sensor, tilt sensor)
3. List the output devices for the Furby. (Hints: speaker, motor for moving parts)
4. Draw a sub-systems diagram for the Furby. Show the system boundary and make clear the difference between real-world inputs/outputs and system sub-systems.
5. Describe how the system works, referring to all sub-systems and how they interact to perform each function.

Section 6: Design Challenges

Engineers use the systems approach to **design** new systems, not just to describe existing ones. Each challenge below gives you a short brief and a numbered specification. Work through parts (a) to (d) in order: start with the big picture, then break it down into sub-systems, then decide how it should be controlled, and finally evaluate your design.

Read every specification point carefully. A good answer shows how **each** point is met — tick them off as you go.

Design Challenge 1: Automatic bike light

A cycling charity wants a rear bike light that riders never forget to switch on. The light should come on by itself when it gets dark, but riders must still be able to switch it off completely when the bike is parked.

Specification — the system must:

- i switch on automatically when the surrounding light level falls below a set level;
- ii switch off automatically when it becomes light again;
- iii have a master on/off switch that the rider can use to disable the system;
- iv flash a red LED so the cyclist is visible to drivers behind.



Design Challenge 1 — Automatic bike light

- (a) Draw a universal system diagram for the bike light. Remember: do not include electricity as an input.
- (b) Draw a sub-systems diagram for the bike light. Show the system boundary, both input devices, the control unit, the output driver and the output device.
- (c) State whether this is an open-loop or closed-loop system. Justify your answer.
- (d) Evaluate your design against the specification. Describe one situation where the light might switch on or off when the rider does not want it to, and suggest an improvement.

Design Challenge 2: Aquarium heater

Tropical fish need their tank water kept at a steady temperature. A pet shop wants a heater unit that holds the water at whatever temperature the owner sets, and warns them if something goes wrong.

Specification — the system must:

- i allow the owner to set the desired water temperature using a dial;
- ii keep the water at the set temperature automatically, switching the heater on and off as needed;
- iii light a warning LED if the water becomes too hot;
- iv be fully enclosed so that no electrical parts touch the water.

Design Challenge 2 — Aquarium heater

- (a) Draw a closed-loop control diagram for the aquarium heater. Label the feedback path and make sure every line has an arrowhead.
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- (b) Name the sub-system that provides feedback and state what information it gives to the control unit.
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- (c) Describe what the system does, step by step, when cold water is added to the tank during cleaning.
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- (d) The pet shop owner suggests a cheaper design that simply switches the heater on for 10 minutes every hour. Explain why the closed-loop design is better for the fish.

Design Challenge 3: Hands-free bin lid

A school canteen wants to reduce the spread of germs by fitting a bin whose lid opens without being touched.

Specification — the system must:

- i open the lid when a hand is detected above the bin;
- ii close the lid automatically a few seconds after the hand is removed;
- iii sound a buzzer when the bin is full;
- iv have a main on/off switch for the canteen staff.

Design Challenge 3 — Hands-free bin lid

- (a) Draw a universal system diagram for the bin. Include any unwanted outputs.
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- (b) Draw a sub-systems diagram for the bin. Show the system boundary and every input device, output driver and output device needed to meet the specification.
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- (c) Describe how the system works, explaining the function of each sub-system in order.
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- (d) Evaluate your design. Identify one part of the system that could fail in a busy canteen and suggest how the design could be improved.